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THE PROBLEM: SAFE SG → GFL TRANSITION

A replacement is local hardware, but stability is a property of the closed grid and its controller architecture.

SG

grid

replace

GFL

port i

$$\delta_i \in \{0, 1\}, \quad S(\delta) = \{i : \delta_i = 1\}$$

$\delta_i = 0$: retain SG _{i} $\delta_i = 1$: replace SG _{i} by GFL _{i}

Engineering objective

$$\max_{S, K} \sum_{i \in S} P_i^0 \quad \text{s.t.} \quad \alpha_{\perp}(S, K) \leq -\sigma_{\text{req}}, \quad K \in \mathcal{K}_n$$

Report the largest independently verified portfolio found.
Certified global optimality is future work.

A safe SG → GFL transition is a portfolio–controller co-design problem.

1

Choose portfolio δ

2

Choose information (architecture)

3

Design gains K_p, K_I

4

Verify grid spectrum + events

2

MODEL: DAE → TRANSVERSE STABILITY

3

IEEE-39: MINIMAL COLLECTIVE BLOCKER

4

EXACT NETWORK CLOSURE: LOCAL × COLLECTIVE

5

BEYOND NODAL DAMPING: EFFECTIVE OPERATORS

6

GRID-EMBEDDED PLL GAIN SYNTHESIS

7

INFORMATION RADIUS AND N-HOP CONTROL

8

N-HOP RESULTS: AUTHORITY AND REPLACEMENT

9

DELAYS, CO-DESIGN, AND NONLINEAR CHECKS

MAIN CONTRIBUTIONS

SCOPE OF THE RESULTS

REFERENCES / CONTACT

PRESERVING ONE MODE DOES NOT PROTECT THE GRID.

More information gives more control authority — only the full grid decides whether a larger SG → GFL transition is safe.

RELIABLE GRIDS
STRONGER SOCIETIES

VANCOUVER 2026

CLEAN ENERGY
BRIGHTER TOMORROW